STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-QIS-EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5497-5510 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 5512-5546, the complete author proof (33 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_sdga.json (pb=1345, stmt 5497-5510, proof 5512-5546).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07RB: sources/stacks-sdga/pinned-original-source.tex, source lines 5546–5546. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09J1: sources/stacks-sdga/pinned-original-source.tex, source lines 5546–5546. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FND: sources/stacks-sdga/pinned-original-source.tex, source lines 5546–5546. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRP: sources/stacks-sdga/pinned-original-source.tex, source lines 1872–1906. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. If $\varphi : \mathcal{A} \to \mathcal{B}$ is a homomorphism of differential graded $\mathcal{O}$-algebras which induces an isomorphism on cohomology sheaves, then*

$$D(\mathcal{A}, d) \longrightarrow D(\mathcal{B}, d), \quad \mathcal{M} \longmapsto \mathcal{M} \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{B}$$

is an equivalence of categories.

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Recall that the restriction functor

$$Mod^{dg}(\mathcal{B}, \mathrm{d}) \to Mod^{dg}(\mathcal{A}, \mathrm{d}), \quad \mathcal{N} \mapsto res_\varphi \mathcal{N}$$

is a right adjoint to

$$Mod^{dg}(\mathcal{A}, \mathrm{d}) \to Mod^{dg}(\mathcal{B}, \mathrm{d}), \quad \mathcal{M} \mapsto \mathcal{M} \otimes_{\mathcal{A}} \mathcal{B}$$

See Section 0FRP. Since restriction sends quasi-isomorphisms to quasi-isomorphisms, we see that it trivially has a left derived extension (given by restriction). This functor will be right adjoint to $- \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{B}$ by Derived Categories, Lemma 0FND. The adjunction map

$$\mathcal{M} \to res_\varphi(\mathcal{M} \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{B})$$

is an isomorphism in $D(\mathcal{A}, \mathrm{d})$ by our assumption that $\mathcal{A} \to \mathcal{B}$ is a quasi-isomorphism of (left) differential graded $\mathcal{A}$-modules. In particular, the functor of the lemma is fully faithful, see Categories, Lemma 07RB. It is clear that the kernel of the restriction functor $D(\mathcal{B}, \mathrm{d}) \to D(\mathcal{A}, \mathrm{d})$ is zero. Thus we conclude by Derived Categories, Lemma 09J1. □